EXERCISES IN POISSON GEOMETRY

The suggested problems for the exercise sessions #1 and #2 are marked with an asterisk. The material from the last section will be discussed in lecture IV, but it's possible (and good) to do the problems beforehand.

1. Poisson manifolds

1.1. Poisson bracket and Poisson bivector field.

1.1.1. *Problem.* Let $\{\cdot,\cdot\}$ be a skew-symmetric bilinear form on $C^\infty(M)$, and denote by

$$\mathrm{Jac}(f,g,h) = \{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\}$$

the *Jacobiator*. Show that if $X_f = \{f,\cdot\}$ is a derivation, then $\mathrm{Jac}(f,g,h)$ is a derivation in each of its arguments. Consequently, the value of $\mathrm{Jac}(f,g,h)$ at m depends only on the differentials of f,g,h at m. In particular, if $\dim M = 2$ it follows that $\mathrm{Jac}(f,g,h) = 0$; thus $\{\cdot,\cdot\}$ is a Poisson bracket.

1.1.2. *Problem.* (Poisson pencils.) Let $\{\cdot,\cdot\}_0$ and $\{\cdot,\cdot\}_1$ be two Poisson brackets on M,with the property that the sum

$$\{f,g\}_0 + \{f,g\}_1$$

is again a Poisson bracket. Prove that all linear combinations

$$a_0\{f,g\}_0 + a_1\{f,g\}_1$$

are Poisson brackets.

1.1.3. *Problem.* Let π be a Poisson structure on a vector space V, and π_0 the leading term of its Taylor expansion at 0. Show that π_0 is again a Poisson structure.

1.1.4. *Problem.* $(*)$ Classify the translation invariant (i.e., 'constant') Poisson structures on a vector space V.

1.1.5. *Problem.* $(*)$ Let π be a Poisson structure on a manifold M, and x_0 a point with $\pi|_{x_0} = 0$. Show that the cotangent space $T^*_{x_0}M$ acquires a Lie bracket.

1.1.6. *Problem.* $(*)$ A vector field X on a Poisson manifold M is called a *Poisson vector field* if it satisfies

$$L_X\{f,g\} = \{L_X f,\, g\} + \{f, L_X g\}$$

for all f,g. In particular, all Hamiltonian vector fields are Poisson vector fields. a) Give an example, as simple as possible, of a Poisson manifold and a Poisson vector field that is not a Hamiltonian vector field.

b) Show that the Hamiltonian vector field are a Lie algebra ideal in the Lie algebra of Poisson vector fields.

1.1.7. *Problem.* Let G be a connected Lie group acting on a symplectic manifold (M,ω), preserving the symplectic structure. A map $\Phi\colon Q \to \mathfrak{g}^*$ is called a *moment map* for the action if it is G-equivariant, and the generating vector fields satisfy

$$\iota(\xi_M)\omega = -\mathrm{d}\langle\Phi,\xi\rangle$$

for all $\xi \in \mathfrak{g}$. Show that Φ is a Poisson map with respect to the Lie-Poisson structure on $\mathfrak{g}^*$. Conversely, show that any Poisson map $\Phi\colon M \to \mathfrak{g}^*$ generates a (local) G-action having Φ as its moment map.

1.1.8. *Problem.* Given a Poisson manifold M, and a free and proper action of a Lie group G by Poisson diffeomorphisms, show that the quotient space M/G inherits a Poisson structure. As an example, show that the Kirillov Poisson structure on $\mathfrak{g}^*$ may be obtained from a G-action on the symplectic manifold T^*G.

1.2. **Submanifolds.** Suppose (M,π) is a Poisson manifold. A submanifold $N \subset M$ is called a *coisotropic submanifold* if it has the property

$$\pi^\sharp(\mathrm{ann}(TN)) \subset TN.$$

It is called a *Poisson submanifold* if $\pi^\sharp(T^*M|_N) \subset TN$.

1.2.1. *Problem.* $(*)$ Let (M,π) be a Poisson manifold.

(a) Show that N is coisotropic, if for all $f,g \in C^\infty(M)$ vanishing on N, the restriction $\{f,g\}|_N$ vanishes on N.
(b) Show that $N \subset M$ is a Poisson submanifold if and only if for all $f,g \in C^\infty(M)$, the restriction $\{f,g\}|_N$ depends only on $f|_N,\ g|_N$.

1.2.2. *Problem.* $(*)$ Show that for any coisotropic submanifold $N \subset M$, the conormal bundle

$$\nu(M,N)^* \cong \mathrm{ann}(TN) \subset T^*M|_N$$

acquires a Lie algebroid structure. *Remark: This problem becomes easier once you know about the Lie algebroid structure on T^*M, for a Poisson manifold M – cf. Lecture II.* What happens for a Poisson submanifold?

1.3. **The Schouten bracket.**

1.3.1. *Problem.* Let $\mathfrak{g}$ be a Lie algebra, and $\wedge^\bullet\mathfrak{g}$ its exterior algebra. Prove the following

Theorem 1.1 (Schouten bracket)**.** *There is a unique structure of a $\mathbb{Z}$-graded super Lie algebra on $\widetilde{\mathfrak{g}}^\bullet := \wedge^{\bullet-1}\mathfrak{g}$, such that*

(i) *The bracket extends that on $\widetilde{\mathfrak{g}}^0 = \mathfrak{g}$;*
(ii) *For fixed $u \in \wedge^k(M)$, the bracket $[u,\cdot]$ is a graded derivation of the wedge product on $\wedge^\bullet\mathfrak{g}$, of degree $k-1$.*

Here, 'super Lie algebra' means that the Jacobi identity and skew-symmetry hold with signs. Thus, $[u,v] = -(-1)^{|u||v|}[v,u]$ and

$$[u,[v,w]] + (-1)^{|u|(|v|+|w|)}[v,[w,u]] + (-1)^{|w|(|u|+|v|)}[w,[u,v]] = 0$$

for homogeneous elements u,v,w.

1.3.2. *Problem.* Let $\mathfrak{X}^\bullet(M) = \Gamma(\wedge^\bullet TM)$ be the $\mathbb{Z}$-graded vector space of multi-vector fields. Prove the following

Theorem 1.2 (Schouten bracket)**.** *There is a unique structure of a $\mathbb{Z}$-graded super Lie algebra on $\mathfrak{k}^\bullet := \mathfrak{X}^{\bullet-1}(M)$, with the following properties:*

(i) *for fixed $u \in \mathfrak{X}^k(M)$, the bracket $[u, \cdot]$ is a graded derivation of degree $k-1$, with respect to the wedge product on $\mathfrak{X}^\bullet(M)$.*
(ii) *For $X \in \mathfrak{X}^1(M)$, the Schouten bracket $[X, \cdot]$ is the Lie derivative.*

Note that the grading for the graded Lie algebra structure on multi-vector fields differs from the grading for the graded algebra structure! For example, $[\cdot, \cdot]$ is a bilinear map from $\mathfrak{X}^k(M) \times \mathfrak{X}^l(M)$ to $\mathfrak{X}^{k+l-1}(M)$, while the wedge product is a bilinear map $\mathfrak{X}^k(M) \times \mathfrak{X}^l(M)$ to $\mathfrak{X}^{k+l}(M)$. Property (ii) above means that

$$[u, v \wedge w] = [u, v] \wedge w + (-1)^{(k-1)l} v \wedge [u, w]$$

if u, v, w are multivector fields of degrees k, l, m.

1.3.3. *Problem.* Using the defining properties of the Schouten bracket, show that

$$[f, u] = -\iota(\mathrm{d}f)u$$

for all $f \in C^\infty(M)$ and $u \in \mathfrak{X}^\bullet(M)$.

1.3.4. *Problem.* Given a bivector field $\pi \in \mathfrak{X}^2(M)$, define $\{f, g\} = \pi(\mathrm{d}f, \mathrm{d}g)$. Show that

$$[\pi, \pi](\mathrm{d}f, \mathrm{d}g, \mathrm{d}h) = -2\,\mathrm{Jac}(f, g, h).$$

1.3.5. *Problem.* From the properties of the Schouten bracket, it follows that

$$\mathrm{d}_\pi := [\pi, \cdot]$$

is a differential on the $\mathbb{Z}$-graded superalgebra $\mathfrak{X}^\bullet(M)$: That is,

$$\mathrm{d}_\pi^2 = \frac{1}{2}[\mathrm{d}_\pi, \mathrm{d}_\pi] = \frac{1}{2}[[\pi, \pi], \cdot] = 0.$$

(a) Give a geometric interpretation to the cocycles in degree 1 and 2, as well as the coboundaries in degree 1.
(b) Show that the map $\pi^\sharp \colon \Omega(M) \to \mathfrak{X}(M)$ is a cochain map.

1.4. **Poisson Lie groups.** A *Poisson Lie group* is a Lie group G, together with a Poisson structure π such that the group multiplication is Poisson. A G-action on a Poisson manifold is called Poisson if the action map $G \times M \to M$ is a Poisson map.

1.4.1. *Problem.* $(*)$ Show that for any Poisson Lie group G, the Poisson structure vanishes at the group unit e.

1.4.2. *Problem.* Show that for a Poisson Lie group G, the inversion map $\mathrm{Inv} \colon G \to G$ is *anti-Poisson*. That is, it becomes a Poisson map $(G, \pi_G) \to (G, -\pi_G)$.

1.4.3. *Problem.* $(*)$ Let G be a Poisson Lie group with a Poisson action on a Poisson manifold M. Show that the space $C^\infty(M)^G$ of invariant functions is closed under the Poisson bracket.

1.5. **Tangent lifts.** Given a vector field $X \in \mathfrak{X}(M)$, let

$$X_T \in \mathfrak{X}(TM)$$

denote its *tangent lift*. That is, the (local) flow of X is obtained by applying the tangent functor to the flow of X. The *vertical lift* of X, denoted $X_V \in \mathfrak{X}(TM)$, is the vertical vector field whose restriction to each fiber T_mM is given by X_m, regarded as a constant vector field. For $f \in C^\infty(M)$, let its tangent lift $f_T \in C^\infty(TM)$ be the function such that

$$f_T(v) = v(f)$$

for all $v \in TM$. We define the vertical lift $f_V \in C^\infty(TM)$ to be simply the pullback.

1.5.1. *Problem.*

(a) Express X_T and f_T in local coordinates.
(b) Show that $X \mapsto X_T$ is a Lie algebra morphism $\mathfrak{X}(M) \to \mathfrak{X}(TM)$.
(c) Show that $(X(f))_T = X_T(f_T)$ for all $X \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$.
(d) Give formulas for $(fg)_T$, $(fX)_T$, for all $X \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$.
(e) Extend the tangent lift operation to a morphism of graded Lie algebras, relative to the Schouten bracket,

$$\mathfrak{X}^{\bullet-1}(M) \to \mathfrak{X}^{\bullet-1}(TM), \quad u \mapsto u_T.$$

Express this lift operation in local coordinates.

1.5.2. *Problem.* As application of the previous problem, every Poisson structure $\pi \in \mathfrak{X}^2(M)$ has a tangent lift to a Poisson structure $\pi_T \in \mathfrak{X}^2(TM)$.

(a) Given a local coordinate expression for π, find the corresponding local coordinate expression for π_T.
(b) Show that if π is nondegenerate, then so is π_T.
(c) More generally, show that if $N \subset M$ is a symplectic leaf of π, then $TN \subset TM$ is a symplectic leaf of π_T.

1.5.3. *Problem.* As we saw, the dual bundle E^* of any Lie algebroid $E \to M$ inherits a Poisson structure. If M is a Poisson manifold, and $E = T^*M$ its cotangent Lie algebroid, show that this Poisson structure on $E^* = TM$ is the tangent lift.

1.5.4. *Problem.* Given a Lie algebroid E over M, construct a 'tangent lift' Lie algebroid E_T over TM.

2. Dirac manifolds

2.1. **Dirac manifolds.**

2.1.1. *Problem.* $(*)$ Prove that for any closed 2-form $\omega \in \Omega^2(M)$, with corresponding map $\omega^\flat \colon TM \to T^*M,\ v \mapsto \omega(v, \cdot)$, the graph

$$E = \mathrm{Gr}(\omega) = \{v + \omega^\flat(v) |\ v \in TM\}$$

is a Dirac structure, with $E \cap T^*M = 0$. Conversely, every Dirac structure E with $E \cap T^*M = 0$ is the graph of a closed 2-form.

2.1.2. *Problem.* $(*)$ Describe a Dirac structure $E \subset \mathbb{T}M$ associated to any (regular) foliation $\mathcal{F}$ of M.

2.1.3. *Problem.* $(*)$ Show that a smooth map $\Phi\colon M' \to M$ between Poisson manifolds is a Poisson map if and only if the pull-back map on 1-forms $\Omega^1(M) \to \Omega^1(M')$ is a Lie algebra homomorphism.

2.1.4. *Problem.* $(*)$ Let $E \subset \mathbb{T}M$ be a Dirac structure, with the property that $E_m = T^*_m M$ for some $m \in M$. Show that the Lie bracket on $\Gamma(E)$ induces a Lie algebra structure on $T^*_m M$.

2.1.5. *Problem.* (Linear Dirac geometry) A *metrized vector space* is a vector space V with a non-degenerate metric. A subspace $E \subset V$ is *Lagrangian*, is $E = E^\perp$. Let $\overline{V}$ be the same vector space with the opposite metric. A *Lagrangian relation* between metrized vector spaces V and V' is a Lagrangian subspaces $L \subset V' \times \overline{V}$. Write $v \sim_L v'$ if $(v', v) \in L$. We define $\ker(L)$ to be the set of all v such that $v \sim_L 0$.

(a) Given a Lagrangian subspace $E \subset V$, show that its *forward image* $E' \subset V'$, consisting of all $v' \in V'$ such that $v \sim_L v'$ for some $v \in E$, is Lagrangian.
(b) Similarly, for a Lagrangian $F' \subset V'$, define its backward image and show that it is Lagrangian.
(c) Suppose that in a) we have $\ker(L) \cap E = 0$, and let E' be the forward image. If $F' \subset V'$ with $E' \cap F' = 0$, show that the backward image $F \subset V$ satisfies $E \cap F = 0$.

2.1.6. *Problem.* $(*)$ Let $\Phi \in C^\infty(M, N)$. Sections $\sigma = X + \alpha \in \Gamma(\mathbb{T}M)$ and $\tau = Y + \beta \in \Gamma(\mathbb{T}N)$ are called Φ-related if

$$\sigma \sim_\Phi \tau :\Leftrightarrow \quad X \sim_\Phi Y, \quad \alpha = \Phi^* \beta.$$

Show that

$$\sigma_1 \sim_\Phi \tau_1, \quad \sigma_2 \sim_\Phi \tau_2 \Rightarrow [\![\sigma_1, \sigma_2]\!] \sim_\Phi [\![\tau_1, \tau_2]\!].$$

2.1.7. *Problem.* Show that the integrability of Lagrangian subbundles $E \subset \mathbb{T}M$ is measured by a certain 3-tensor $\Upsilon_E \in \Gamma(\wedge^3 E^*)$.

2.1.8. *Problem.* (Mackenzie-Xu) Let $E, F \subset \mathbb{T}M$ be two Dirac structures, with $E \cap F = 0$. Let $e_1, \dots, e_n$ be a (local) basis of sections of E, and $f^1, \dots, f^n$ the dual basis of sections of F. Letting $\mathsf{a}\colon \mathbb{T}M \to TM$ be the projection to the vector field part, show that

$$\pi = \sum_i \mathsf{a}(e_i) \wedge \mathsf{a}(f^i)$$

is a Poisson structure.

3. Symplectic groupoids

3.1. **Lie groupoid cochains.** Let $\mathcal{G} \rightrightarrows M$ be a Lie groupoid, and $\mathcal{G}^{(k)} \subset \mathcal{G}^k$ its space of k-arrows. Define maps $\partial_i\colon \mathcal{G}^{(k+1)} \to \mathcal{G}^{(k)},\ i = 0, \dots, k+1$ by

$$\partial_i(g_1, \dots . g_{k+1}) = (g_1, \dots, g_i \circ g_{i+1}, \dots, g_{k+1})$$

for $0 < i < k+1$, while the map ∂_0 drops g_1 and ∂_{k+1} drops g_{k+1}.[1] Let

$$\delta = \sum_{i=0}^{k+1} (-1)^i \partial_i^* \colon \ \Omega^\bullet(\mathcal{G}^{(k)}) \to \Omega^\bullet(\mathcal{G}^{(k+1)}).$$

Then $\delta \circ \delta = 0$. (Exercise.)

3.1.1. *Problem.* A form $\alpha \in \Omega^l(\mathcal{G})$ is called *multiplicative* if it satisfies $\delta\alpha = 0$. For $l = 0$ this means, $f(g_1 \circ g_2) = f(g_1) + f(g_2)$. Show that a symplectic 2-form ω on a groupoid $\mathcal{G} \rightrightarrows M$ is multiplicative if and only if the graph of the groupoid multiplication is a Lagrangian submanifold $\mathrm{Gr}(\mathrm{Mult}_{\mathcal{G}}) \subset \mathcal{G} \times \overline{\mathcal{G} \times \mathcal{G}}$.

3.2. **Bisections.** An embedded submanifold $S \subset \mathcal{G}$ of a Lie groupoid $\mathcal{G} \rightrightarrows M$ is a *bisection* if both $\mathtt{s}$ and $\mathtt{t}$ restrict to diffeomorphisms $S \to M$. (Thus, S can be regarded as a section of $\mathtt{s}$, and also of $\mathtt{t}$.) Let $\Gamma(\mathcal{G})$ be the set of all bisections.

3.2.1. *Problem.*

(a) Show that $\Gamma(\mathcal{G})$ is naturally a group, and define actions $\Gamma(\mathcal{G}) \to \mathrm{Diff}(\mathcal{G})$ of left multiplication, right multiplication, and conjugation (adjoint action) on $\mathcal{G}$. Show that the conjugation action is by automorphisms of the groupoid structure.

(b) Describe the group $\Gamma(\mathcal{G})$, and its actions, for the following basic examples:
   (i) $\mathcal{G} = G \rightrightarrows \mathrm{pt}$ a Lie group,
   (ii) $\mathcal{G} = M \times \mathcal{M} \rightrightarrows M$ a pair groupoid,
   (iii) $\mathcal{G} \cong G \times M \rightrightarrows M$ an action groupoid for a G-action on M,
   (iv) $\mathcal{G} \rightrightarrows M$ the gauge groupoid of a principal G-bundle $P \to M$.

(c) For a symplectic groupoid $(\mathcal{G}, \omega)$, show that the actions above preserve the symplectic form if and only if S is a Lagrangian submanfold. Show that the Lagrangian bisections form a subgroup $\Gamma_{Lag}(\mathcal{G}) \subset \Gamma(\mathcal{G})$.

3.3. **Symplectic realization.** Let (M, π) be a Poisson manifold. A *symplectic realization* of (M, π) is a Poisson manifold (P, π_P), with π_P non-degenerate (i.e. symplectic), together with a Poisson map $P \to M$. It is called *full* if it is a surjective submersion. An important example is $M = \mathfrak{g}^*$ with the standard Lie-Poisson structure: a full symplectic realization is given by $P = T^*G$, where G is a Lie group integrating $\mathfrak{g}$, and with the map

$$T^*G \to \mathfrak{g}^*$$

given by left trivialization.

3.3.1. *Problem.* $(*)$ (Following Cannas da Silva-Weinstein) Let $M = \mathbb{R}^2$ with the Poisson structure

$$\pi = x_1 \frac{\partial}{\partial x_1} \wedge \frac{\partial}{\partial x_2}.$$

Let $P = \mathbb{R}^4$, with coordinates q_1, q_2, p_1, p_2, with the Poisson structure

$$\pi_P = \frac{\partial}{\partial q_1} \wedge \frac{\partial}{\partial p_1} + \frac{\partial}{\partial q_2} \wedge \frac{\partial}{\partial p_2}$$

[1] Intuitively, a k-arrow involves $k+1$ base points $x_0, \ldots, x_k$, and arrows g_i going from x_i to x_{i-1}. The map ∂_i is such that it omits the i-th base point.

corresponding to the standard symplectic form. Let $i\colon M \to P$ be the inclusion $(x_1, x_2) \mapsto (x_1, x_2, 0, 0)$.

a) Show that the map

$$\mathtt{t}\colon P \to M, \quad (q_1, p_1, q_2, p_2) \mapsto (q_1,\ q_2 + p_1 q_1)$$

is Poisson. Thus, $\mathtt{t}$ defines a symplectic realization, with $\mathtt{t} \circ i = \mathrm{id}_M$.

b) Find a submersion $\mathtt{s}\colon P \to M$, with $\mathtt{s} \circ i = \mathrm{id}_M$, such that the fibers are symplectically orthogonal to those of $\mathtt{t}$. Verify that $\mathtt{s}$ is anti-Poisson.

c) Not using the formulas from parts a) and b), argue that M admits a global symplectic groupoid integrating it. (Hint: This Poisson structure on $\mathbb{R}^2$ is a special case of ...).

3.3.2. *Problem.* $(*)$ Find a full symplectic realization for any Poisson structure of the form

$$f(q_1, q_2) \frac{\partial}{\partial q_1} \wedge \frac{\partial}{\partial q_2},$$

on $\mathbb{R}^2$, where f is a smooth function.

3.3.3. *Problem.* $(*)$ Consider $\pi = \frac{\partial}{\partial q_1} \wedge \frac{\partial}{\partial q_2}$ on $\mathbb{R}^2$. Using the Poisson spray

$$X = p_1 \frac{\partial}{\partial q_2} - p_2 \frac{\partial}{\partial q_1},$$

on $P = T^*\mathbb{R}^2$, show that the Poisson structure from Crainic-Marcut's symplectic realization is

$$\pi_P = \frac{\partial}{\partial q_1} \wedge \frac{\partial}{\partial p_1} + \frac{\partial}{\partial q_2} \wedge \frac{\partial}{\partial p_2} - \frac{\partial}{\partial q_1} \wedge \frac{\partial}{\partial q_2},$$

with $\mathtt{s}(q, p) = q$ and $\mathtt{t}(q, p) = (q_1 - p_2, q_2 + p_1)$.

3.3.4. *Problem.* As a more interesting example, work out the Crainic-Marcut construction for $\mathfrak{g}^*$, the dual of a Lie algebra. (The result will be $T^*G \cong \mathfrak{g}^* \times G$, in 'exponential' coordinates' on G near e.)